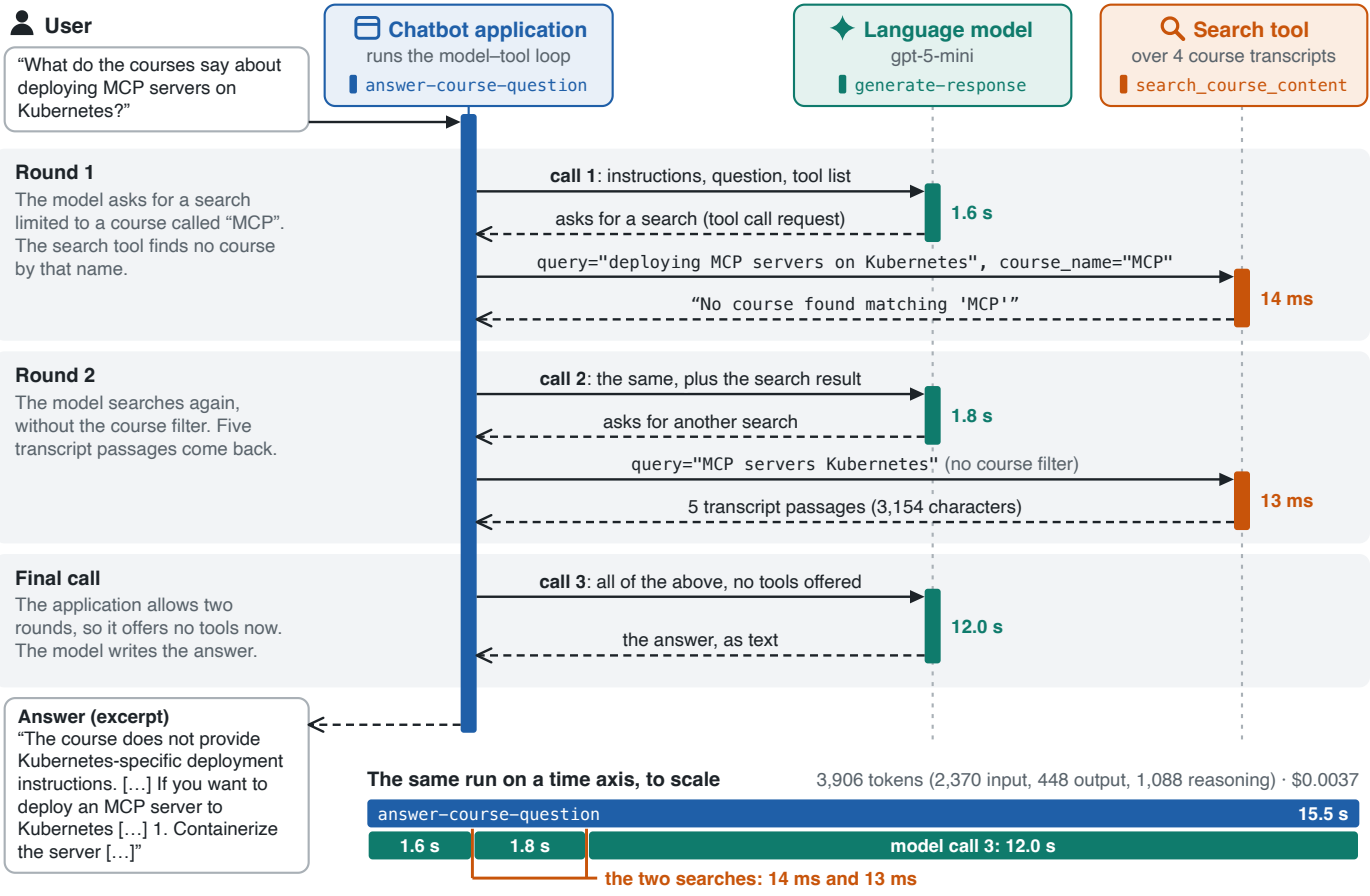

User
“What do the courses say about deploying MCP servers on Kubernetes?”
Chatbot application
runs the model–tool loop
answer_course_question
Language model
gpt-5-mini
generate_response
Search tool
over 4 course transcripts
search_course_content
Round 1
The model asks for a search limited to a course called “MCP”. The search tool finds no course by that name.
call 1: instructions, question, tool list
asks for a search (tool call request)
1.6 s
query="deploying MCP servers on Kubernetes", course_name="MCP"
“No course found matching 'MCP'”
14 ms
Round 2
The model searches again, without the course filter. Five transcript passages come back.
call 2: the same, plus the search result
asks for another search
1.8 s
query="MCP servers Kubernetes" (no course filter)
5 transcript passages (3,154 characters)
13 ms
Final call
The application allows two rounds, so it offers no tools now. The model writes the answer.
call 3: all of the above, no tools offered
the answer, as text
12.0 s
Answer (excerpt)
“The course does not provide Kubernetes-specific deployment instructions. […] If you want to deploy an MCP server to Kubernetes […] 1. Containerize the server […]”
The same run on a time axis, to scale
3,906 tokens (2,370 input, 448 output, 1,088 reasoning) · $0.0037
answer_course_question
15.5 s
1.6 s
1.8 s
model call 3: 12.0 s
the two searches: 14 ms and 13 ms